

Technical Document #113: Data Engineering - Comprehensive Overview

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Data lakes store raw data in its native format. They support structured, semi-structured, and unstructured data at any scale.

Partitioning divides data into smaller, more manageable pieces. It improves query performance by reducing the amount of data scanned.

Data warehouses store structured data optimized for analytical queries. They use columnar storage and support complex aggregations.

Data pipelines automate the movement and transformation of data between systems. Apache Airflow and Prefect are popular workflow orchestration tools.

ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a target data store.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Key Takeaways:

- Data governance establishes policies and procedures for managing data assets.
- Data lakes store raw data in its native format.
- Stream processing handles continuous data streams in real-time.